

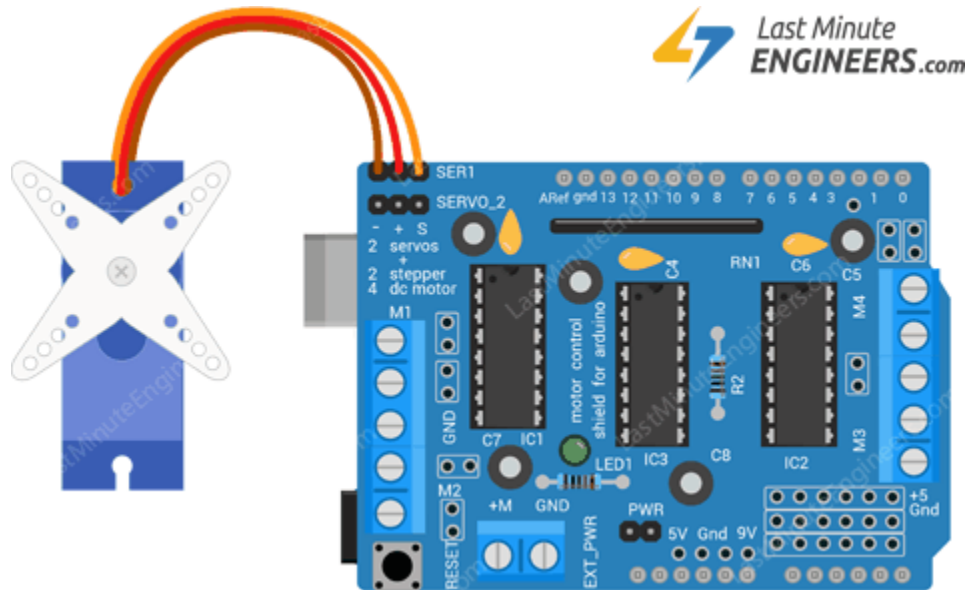
## Driving Servo Motors with L293D Shield



The motor shield actually breaks out Arduino's 16bit PWM output pins **#9 & #10** to the edge of the shield with two 3-pin headers.

Power for the Servos comes from the Arduino's on-board 5V regulator, so you don't have to connect anything to the EXT\_PWR terminal.

## Wiring Servo Motor to L293D Motor Shield & Arduino



As we are using the onboard PWM pins, the sketch uses IDE's built in Servo library.

## Program

```
#include <Servo.h>

Servo myservo; // create servo object to control a servo

int pos = 0;    // variable to store the servo position

void setup()
{
    // attaches the servo on pin 10 to the servo object
    myservo.attach(10);
}

void loop()
{
    // sweeps from 0 degrees to 180 degrees
    for(pos = 0; pos <= 180; pos += 1)
    {
        myservo.write(pos);
        delay(15);
    }
    // sweeps from 180 degrees to 0 degrees
    for(pos = 180; pos >= 0; pos -= 1)
    {
        myservo.write(pos);
        delay(15);
    }
}
```

### Program for servo motor Without shield

```
#include <Servo.h>

int servoPin = 9;

Servo servo;

int angle = 0;// servo position in degrees

void setup()
{
  servo.attach(servoPin);
}

void loop()
{
  //scan from 0 to 180 degrees
  for(angle = 0; angle<180; angle++)
  {
    servo.write(angle);
    delay(20);
  }
  //now scan back from 180 to 0 degrees
  for(angle = 180;angle>0;angle--)
  {
    servo.write(angle);
    delay(20);
  }
}
```